CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

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1. Objective

The objective of this project is to develop a machine learning model to identify fraudulent credit card transactions. The model helps distinguish between genuine and fraudulent transactions using transaction data.

2. Dataset Description

The dataset contains credit card transaction records. Each transaction is labeled as:

- Class 0: Genuine Transaction
- Class 1: Fraudulent Transaction

The dataset contains several features including Time, Amount, and anonymized variables (V1 to V28).

3. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the credit card transaction dataset.
2. Checked for missing values.
3. Normalized the Time and Amount columns using StandardScaler.
4. Separated the dataset into features (X) and target variable (y).

4. Handling Class Imbalance

The dataset was highly imbalanced because fraudulent transactions were much fewer than genuine transactions.

To address this issue, SMOTE (Synthetic Minority Oversampling Technique) was applied to generate synthetic fraud samples and balance the dataset.

5. Model Training:

A Logistic Regression classification model was used for training.

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

The model was trained using the training dataset and evaluated using the testing dataset.

6. Model Evaluation

The performance of the model was evaluated using the following metrics:

- Precision
- Recall
- F1-Score
- Accuracy

Results

Metric	Value
Accuracy	98%
Precision	98% - 99%
Recall	98% - 99%

F1-Score	98%

Classification Report:

Class 0 (Genuine Transactions)

- Precision: 0.98
- Recall: 0.99
- F1-Score: 0.98

Class 1 (Fraudulent Transactions)

- Precision: 0.99
- Recall: 0.98
- F1-Score: 0.98

7. Conclusion

A machine learning model was successfully developed to detect fraudulent credit card transactions. Data preprocessing and normalization techniques were applied, and SMOTE was used to handle class imbalance.

The Logistic Regression model achieved an accuracy of 98% with high precision, recall, and F1-score values. The results indicate that the model effectively identifies fraudulent transactions and can be used as a reliable fraud detection system.

8. Tools and Technologies Used

- Python
- Google Colab
- Pandas
- Scikit-learn
- Imbalanced-learn (SMOTE)
- Logistic Regression